

Mathematical Logic

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Preposition (or) Statement :

(21)

A preposition (or) statement is a declarative sentence that either true or false but not both.

Eg: 1. Sun rises in the west (statement)

2. $2 + 3 = 5$ (statement)

3. $4 + x = 7$ (Not a statement)

4. $6 + 3 = 10$ (statement)

5. Close the door (Command)

6. Read the book carefully (Command)

7. What is the time? (Question)

8. What a cool day (Exclamation)

Connectives

1. Name : Negation

Representation : $\sim, \neg$

Meaning : Not

Truth table :

p	$\sim p$
T	F
F	T

Ex: p : London is in England

$\sim p$: London is not in England

2. Name : Conjunction

Representation : $\wedge$ (cap, meet)

Meaning : And

Truth table :

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Ex: p : Ramu is studying very well.

q : His father will buy a bike.

$p \wedge q$: Ramu is studying very well and his father will buy a bike.

3. Name : Disjunction

Representation : $\vee$ (cup, join)

Meaning : OR

Truth table :

P	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Ex: P : It is cold

q : It is raining

$P \vee q$: It is cold or it is raining

4. Name : Conditional

Representation : $\rightarrow, \Rightarrow$

Meaning : If then (Implication)

Truth table :

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Ex: P : Today is Thursday

q : Tomorrow is Friday

$P \rightarrow q$: If today is Thursday then tomorrow is Friday

NOTE : $P \rightarrow q$

Antecedent
(Hypothesis)

consequent
(conclusion)

5. Name : Bi-conditional

Representation : $\leftrightarrow, \Leftrightarrow, \Rightarrow, \equiv$

Meaning : If and only if equivalence

Truth table :

P	q	$P \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Ex: p : He swims

q : The water is warm

$p \leftrightarrow q$: He swims if and only if the water is warm

Construction of Truth tables:

1. $[q \wedge (p \rightarrow q)] \rightarrow p$
2. $\sim [p \vee (q \wedge r)] \leftrightarrow (p \wedge q) \vee (p \vee r)$
3. $(p \vee q) \wedge q$
4. $(p \vee q) \vee \sim p$
5. $(p \vee \sim q) \rightarrow p$
6. $[\sim (p \wedge q) \vee r] \rightarrow \sim p$
7. $(p \rightarrow q) \vee (\sim p \rightarrow q)$
8. $(p \rightarrow q) \rightarrow r$
9. $p \rightarrow (q \rightarrow r)$
10. $\sim (p \vee \sim q) \rightarrow \sim p$

1. $[q \wedge (p \rightarrow q)] \rightarrow p$

P	q	$p \rightarrow q$	$q \wedge (p \rightarrow q)$	$[q \wedge (p \rightarrow q)] \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

2. $\sim [p \vee (q \wedge r)] \leftrightarrow (p \wedge q) \vee (p \vee r)$

P	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$\sim [p \vee (q \wedge r)]$ A	$p \wedge q$	$p \vee r$	$(p \wedge q) \wedge (p \vee r)$ B	$A \rightarrow B$
T	T	T	T	T	F	T	T	T	F
T	T	F	F	T	F	T	T	T	F
T	F	T	F	T	F	F	T	F	T
T	F	F	F	T	F	F	T	F	T
F	T	T	T	T	F	F	T	F	T
F	T	F	F	F	T	F	F	F	F
F	F	T	F	F	T	F	T	F	F
F	F	F	F	F	T	F	F	F	F

$$3. (p \vee q) \wedge q$$

p	q	$p \vee q$	$(p \vee q) \wedge q$
T	T	T	T
T	F	T	F
F	T	T	T
F	F	F	F

$$4. (p \vee q) \vee \sim p$$

p	q	$p \vee q$	$\sim p$	$(p \vee q) \vee \sim p$
T	T	T	F	T
T	F	T	F	T
F	T	T	T	T
F	F	F	T	T

$$5. (p \vee \sim q) \rightarrow p$$

p	q	$\sim q$	$p \vee \sim q$	$(p \vee \sim q) \rightarrow p$
T	T	F	T	T
T	F	T	T	T
F	T	F	F	T
F	F	T	T	F

$$8. (p \rightarrow q) \rightarrow r$$

p	q	r	$p \rightarrow q$	$(p \rightarrow q) \rightarrow r$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	T
F	T	T	T	T
F	T	F	T	F
F	F	T	T	T
F	F	F	T	F

$$6. [\sim(p \wedge q) \vee r] \rightarrow \sim p$$

p	q	r	$p \wedge q$	$\sim(p \wedge q)$	$\sim(p \wedge q) \vee r$	$\sim p$	$[\sim(p \wedge q) \vee r] \rightarrow \sim p$
T	T	T	T	F	F	F	F
T	T	F	T	F	F	F	T
T	F	T	F	T	T	F	F
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	T
F	T	F	F	T	T	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

$$7. (p \rightarrow q) \vee (\sim p \rightarrow q)$$

p	q	$p \rightarrow q$	$\sim p$	$\sim p \rightarrow q$	$(p \rightarrow q) \vee (\sim p \rightarrow q)$
T	T	T	F	F	T
T	F	F	F	T	T
F	T	T	T	T	T
F	F	T	T	T	T

9. $p \rightarrow (q \rightarrow r)$

p	q	r	$q \rightarrow r$	$p \rightarrow (q \rightarrow r)$
T	T	T	T	T
T	T	F	F	F
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	T
F	F	T	T	T
F	F	F	T	T

10. $\sim (p \vee \sim q) \rightarrow \sim p$

p	q	$\sim q$	$p \vee \sim q$	$\sim (p \vee \sim q)$	$\sim p$	$\sim (p \vee \sim q) \rightarrow \sim p$
T	T	F	T	F	F	T
T	F	T	T	F	F	T
F	T	F	F	T	T	T
F	F	T	T	F	T	T

Tautology	Contradiction	Contingency
The given statement is True	The given statement is False	The given statement is both combination of True and False

1. Using Truth table to show that $(p \wedge (p \leftrightarrow q)) \rightarrow q$ is a tautology.

p	q	$p \leftrightarrow q$	$p \wedge (p \leftrightarrow q)$	$(p \wedge (p \leftrightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	F	F	T
F	F	T	F	T

$\therefore (p \wedge (p \leftrightarrow q)) \rightarrow q$ is a tautology

2. show that $[(p \vee q) \vee \sim p]$ is a tautology

3. $(q \vee (p \wedge \sim q)) \vee (\sim p \wedge \sim q)$

4. $[(p \wedge \sim q) \rightarrow r] \rightarrow [p \rightarrow (q \vee r)]$

5. $[(p \rightarrow q) \rightarrow r] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$

2. $(p \vee q) \vee \sim p$

p	q	$p \vee q$	$\sim p$	$(p \vee q) \vee \sim p$
T	T	T	F	T
T	F	T	F	T
F	T	T	T	T
F	F	F	T	T

$\therefore (p \vee q) \vee \sim p$ is a tautology

3. $[q \vee (p \wedge \sim q)] \vee (\sim p \wedge \sim q)$

p	q	$\sim p$	$\sim q$	$p \wedge \sim q$	$q \vee (p \wedge \sim q)$	$\sim p \wedge \sim q$	$A \vee B$
T	T	F	F	F	T	F	T
T	F	F	T	T	T	F	T
F	T	T	F	F	T	F	T
F	F	T	T	F	F	T	T

$\therefore [q \vee (p \wedge \sim q)] \vee (\sim p \wedge \sim q)$ is a tautology.

4. $[(p \wedge \sim q) \rightarrow r] \rightarrow [p \rightarrow (q \vee r)]$

p	q	r	$\sim q$	$p \wedge \sim q$	$(p \wedge \sim q) \rightarrow r$	$q \vee r$	$p \rightarrow (q \vee r)$	$A \rightarrow B$
					A		B	
T	T	T	F	F	T	T	T	T
T	T	F	F	F	T	T	T	T
T	F	T	T	T	T	T	T	T
T	F	F	T	T	F	F	F	T
F	T	T	F	F	T	T	T	T
F	T	F	F	F	T	T	T	T
F	F	T	T	F	T	T	T	T
F	F	F	T	F	T	F	T	T

$\therefore [(p \wedge \sim q) \rightarrow r] \rightarrow [p \rightarrow (q \vee r)]$ is a tautology

$$6. [(p \rightarrow q) \rightarrow r] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$$

P	q	r	$p \rightarrow q$	$(p \rightarrow q) \rightarrow r$	$p \rightarrow r$	$(p \rightarrow q) \rightarrow (p \rightarrow r)$	$A \rightarrow B$
				A		B	
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	T	T
T	F	F	F	T	F	T	T
F	T	T	T	T	T	T	T
F	T	F	T	F	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	F	T	T	T

$$\therefore [(p \rightarrow q) \rightarrow r] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$$

Contradiction:

1. Using truth table to show $p \wedge \sim p$ is a contradiction

P	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

$\therefore p \wedge \sim p$ is a contradiction

2. Show that $\sim [[(p \vee q) \rightarrow r] \wedge (\sim p)] \rightarrow (q \rightarrow r)$

P	q	r	$p \vee q$	$(p \vee q) \rightarrow r$	$\sim p$	$A \wedge \sim p$	$q \rightarrow r$	$(A \wedge \sim p) \rightarrow (q \rightarrow r)$	$\sim B$
				A				B	
T	T	T	T	T	F	F	T	T	F
T	T	F	T	F	F	F	F	T	F
T	F	T	T	T	F	F	T	T	F
T	F	F	T	F	F	F	T	T	F
F	T	T	T	T	T	T	T	T	F
F	T	F	T	T	T	T	F	T	F
F	F	T	F	T	T	T	T	T	F
F	F	F	F	T	T	T	T	T	F

$\therefore \sim [[(p \vee q) \rightarrow r] \wedge (\sim p)] \rightarrow (q \rightarrow r)$ is a contradiction

Converse, Contrapositive and Opposite (or) Inverse:
 $\neg(p \rightarrow q) \equiv (p \wedge \neg q)$ $(\neg q \rightarrow \neg p)$ $(\neg p \rightarrow \neg q)$

Ex:

1. * If $\triangle ABC$ is a right angle triangle then $|AC|^2 = |AB|^2 + |BC|^2$

Let P : $\triangle ABC$ is a right angle triangle

$$Q: |AC|^2 = |AB|^2 + |BC|^2$$

Converse: If $|AC|^2 = |AB|^2 + |BC|^2$ then $\triangle ABC$ is a right angle triangle

Contrapositive: If $|AC|^2 \neq |AB|^2 + |BC|^2$ then $\triangle ABC$ is not a right angle triangle

Opposite: If $\triangle ABC$ is not a right angle triangle then $|AC|^2 \neq |AB|^2 + |BC|^2$

2. Write Converse, Contrapositive and Opposite of "If today is a holiday then I will go for a movie"

Let P : If Today is a holiday

Q : I will go for a movie.

Converse: If I will go for a movie then today is a holiday.

Contrapositive: If I will not go for a movie then today is not a holiday.

Opposite: If today is not a holiday then I will not go for a movie.

→ Logically equivalent:

Two statement formulas P and Q are said to be logically equivalent if $p \leftrightarrow q$ is a tautology

Ex: $(p \rightarrow q) \equiv (\neg p \vee q)$

P	Q	$P \rightarrow Q$	$\neg P$	$(\neg P \vee Q)$	$(P \rightarrow Q) \leftrightarrow (\neg P \vee Q)$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

i.e., $(p \rightarrow q) \equiv (\neg p \vee q)$ is a tautology

Equivalent formulas:

1. Idempotent : $p \vee p \equiv p$

$$p \wedge p \equiv p$$

2. Commutative : $p \vee q \equiv q \vee p$

$$p \wedge q \equiv q \wedge p$$

3. Associative : $p \vee (q \vee r) \equiv (p \vee q) \vee r$

$$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$$

4. Identity : $p \vee T \equiv T$, $p \wedge T \equiv p$

$$p \vee F \equiv p$$

5. Demorgan's : $\sim(p \vee q) \equiv (\sim p) \wedge (\sim q)$

$$\sim(p \wedge q) \equiv (\sim p) \vee (\sim q)$$

6. Distributive : $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

7. Absorption : $p \vee (p \wedge q) \equiv p$

$$p \wedge (p \vee q) \equiv p$$

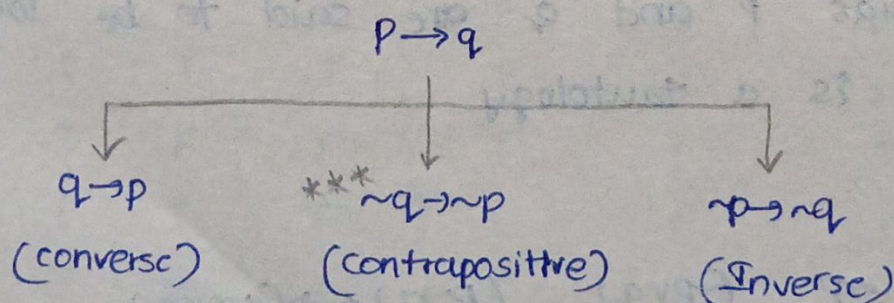
8. Double Negation : $\sim(\sim p) \equiv p$

9. $(p \rightarrow q) \equiv (\sim q) \rightarrow (\sim p)$

10. $(p \rightarrow q) \equiv \sim p \vee q$

11. $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

logical infinitives



9. Domination law : $p \vee T \equiv T$

$$p \wedge F \equiv F$$

10. Negation law : $p \vee \sim p \equiv T$

$$p \wedge \sim p \equiv F$$

NOTE:

1. $p \rightarrow q \equiv \neg p \vee q$

2. $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
 $\equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

Well formed Formula (WFF)

A statement formula is an expression denoted by a string containing variables, ~~characteristics~~ paranthesis and connective symbols

Rules:

1. A Statement variable is standing alone is a wellformed formula
2. If A is a well formed formula then $\neg A$ is also a well-formed formula
3. If A, B are two well-formed formulas then $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, $(A \leftrightarrow B)$ are also well formed formulas

Ex:

1. $\neg(p \wedge q)$ - is a well formed formula
2. $\neg(p \vee \neg q)$ - is a well formed formula
3. $(p \rightarrow (p \wedge q))$ - is a well formed formula
4. $(p \wedge q) \rightarrow \neg p$ - is not a well formed formula
5. $p \rightarrow (q \rightarrow r)$ - is not a well formed formula

Duality law:

Let x, x' are two statement formulas is said to be dual to each other by replacing $\wedge$ by $\vee$ and $\vee$ by $\wedge$, also replacing $\top$ by $\bot$ and $\bot$ by $\top$

Ex: Write the dualities of the following expressions

1. $((p \wedge q) \rightarrow r)$

The duality of the above expression is

$((p \vee q) \rightarrow r)$

2. $((p \rightarrow q) \rightarrow \top)$

The duality of the above expression is

$((p \rightarrow q) \rightarrow \bot)$

Without using Truth tables

1. To eliminate conditional($\rightarrow$), Biconditional($\leftrightarrow$) by using suitable expressions
2. To eliminate $\sim$ before the paranthesis
3. To use demorgan law & distributive laws repeatedly.

1. Show that $p \rightarrow (q \rightarrow r) \Leftrightarrow p \rightarrow (\sim q \vee r) \Leftrightarrow (p \wedge q) \rightarrow r$

sol:

$$\begin{aligned} p \rightarrow (q \rightarrow r) &\Leftrightarrow p \rightarrow (\sim q \vee r) & (\because p \rightarrow q \Leftrightarrow \sim p \vee q) \\ &\Leftrightarrow \sim p \vee (\sim q \vee r) & (\because p \rightarrow q \Leftrightarrow \sim p \vee q) \\ &\Leftrightarrow (\sim p \vee \sim q) \vee r & (\because \text{Associative law}) \\ &\Leftrightarrow \sim(p \wedge q) \vee r & (\because \text{Demorgan's law}) \\ &\Leftrightarrow (p \wedge q) \rightarrow r & (\because p \rightarrow q \Leftrightarrow \sim p \vee q) \end{aligned}$$

2. Show that $(p \rightarrow q) \rightarrow q \Leftrightarrow p \vee q$

sol:

$$\begin{aligned} (p \rightarrow q) \rightarrow q &\Leftrightarrow (\sim p \vee q) \rightarrow q & (\because p \rightarrow q \Leftrightarrow \sim p \vee q) \\ &\Leftrightarrow \sim(\sim p \vee q) \vee q & (\because p \rightarrow q \Leftrightarrow \sim p \vee q) \\ &\Leftrightarrow (\sim(\sim p) \wedge (\sim q)) \vee q & (\because \text{Demorgan's law}) \\ &\Leftrightarrow (p \wedge \sim q) \vee q \\ &\Leftrightarrow (p \vee q) \wedge (\sim q \vee q) & (\because \text{Distributive law}) \\ &\Leftrightarrow (p \vee q) \wedge T & (\because \text{Negation law}) \\ &\Leftrightarrow p \vee q & (\because \text{Identity law}) \end{aligned}$$

3. Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology

sol:

$$\begin{aligned} (p \wedge q) \rightarrow (p \vee q) &\Leftrightarrow \sim(p \wedge q) \vee (p \vee q) & (\because p \rightarrow q \Leftrightarrow \sim p \vee q) \\ &\Leftrightarrow (\sim p \vee \sim q) \vee (p \vee q) & (\because \text{Demorgan's law}) \\ &\Leftrightarrow (\sim p \vee p) \vee (\sim q \vee q) & (\because \text{commutative \& distributive laws}) \\ &\Leftrightarrow T \vee T & (\because \text{Negation law}) \\ &\Leftrightarrow T & (\because \text{idempotent law}) \end{aligned}$$

4. Show that $\sim(p \vee (\sim p \wedge q))$ and $(\sim p \wedge \sim q)$ are logically equivalent

$$\begin{aligned} \text{sol: } \sim(p \vee (\sim p \wedge q)) &\Leftrightarrow \sim p \wedge \sim(\sim p \wedge q) & (\because \text{Demorgan's}) \\ &\Leftrightarrow \sim p \wedge (p \vee \sim q) & (\because \text{Demorgan's}) \\ &\Leftrightarrow (\sim p \wedge p) \vee (\sim p \wedge \sim q) & (\because \text{Distributive}) \\ &\Leftrightarrow F \vee (\sim p \wedge \sim q) & (\because \text{Negation}) \\ &\Leftrightarrow (\sim p \wedge \sim q) & (\because \text{Identity}) \end{aligned}$$

$$\therefore \sim(p \vee (\sim p \wedge q)) \equiv \sim p \wedge \sim q$$

5. Show that $(\sim p \wedge (\sim q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \Leftrightarrow r$

$$\begin{aligned} \text{sol: } (\sim p \wedge (\sim q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) &\Leftrightarrow ((\sim p \wedge \sim q) \wedge r) \vee ((q \vee p) \wedge r) & (\because \text{Associative \& Distributive}) \\ &\Leftrightarrow ((\sim p \wedge \sim q) \vee (q \vee p)) \wedge r & (\because \text{Distributive}) \\ &\Leftrightarrow (\sim(p \wedge q) \vee (p \vee q)) \wedge r & (\because \text{Demorgan's \& commutative}) \\ &\Leftrightarrow T \wedge r & (\because \text{Negation}) \\ &\Leftrightarrow r & (\because \text{Identity}) \end{aligned}$$

6. Show that $((p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$ is a tautology

sol: Given that

$$((p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$$

$$\begin{aligned} \text{Let } A &\Leftrightarrow ((p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))) \\ &\Leftrightarrow ((p \vee q) \wedge (p \vee \sim(\sim q \vee \sim r))) & (\because \text{Demorgan's}) \\ &\Leftrightarrow (p \vee q) \wedge (p \vee (q \wedge r)) & (\because \text{Demorgan's}) \\ &\Leftrightarrow (p \vee q) \wedge (p \vee q) \wedge (p \vee r) & (\because \text{Distributive}) \\ &\Leftrightarrow (p \vee q) \wedge (p \vee r) & (\because \text{Idempotent}) \\ &\Leftrightarrow p \vee (q \wedge r) \end{aligned}$$

$$\begin{aligned} \text{Let } B &\Leftrightarrow (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r) \\ &\Leftrightarrow \sim p \wedge (\sim q \vee \sim r) & (\because \text{Distributive}) \\ &\Leftrightarrow \sim(p \vee (q \wedge r)) & (\because \text{Demorgan's}) \end{aligned}$$

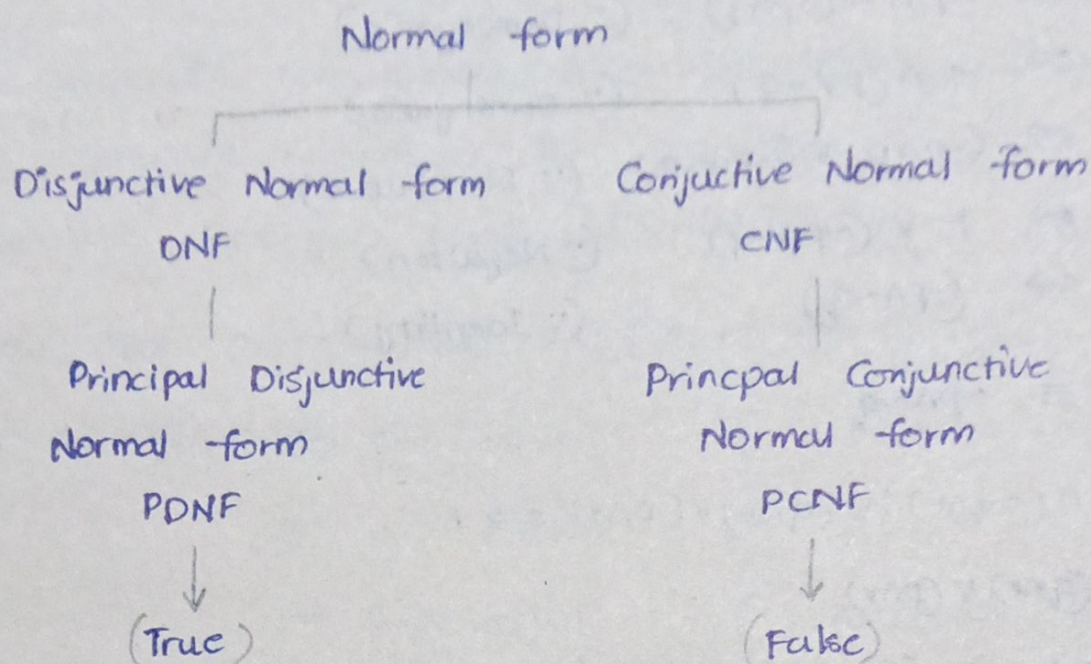
$$\begin{aligned} \text{Now } A \vee B &\Leftrightarrow p \vee (q \wedge r) \vee \sim(p \vee (q \wedge r)) \\ &\Leftrightarrow T & (\because \text{Negation}) \end{aligned}$$

$$\therefore ((p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r) \text{ is a tautology}$$

$$\Leftrightarrow T$$

Hence proved.

Normal forms



- * $\wedge$ - product operator
- * $\vee$ - sum operator

* Min terms

2 variables (p, q)

$p \wedge q$
 $\sim p \wedge q$
 $p \wedge \sim q$
 $\sim p \wedge \sim q$

3 variables (p, q, r)

$p \wedge q \wedge r$
 $\sim p \wedge q \wedge r$
 $p \wedge \sim q \wedge r$
 $p \wedge q \wedge \sim r$
 $\sim p \wedge \sim q \wedge r$
 $\sim p \wedge q \wedge \sim r$
 $p \wedge \sim q \wedge \sim r$
 $\sim p \wedge \sim q \wedge \sim r$

PDNF : A statement formula which is equivalent to given formula consisting of disjunction of min terms only. is called PDNF (or) sum of products canonical form.

1. Find the PDNF of $\sim p \vee q$

$$\begin{aligned}
 \text{sol: } \sim p \vee q &\Leftrightarrow (\sim p \wedge T) \vee (q \wedge T) && (\because \text{Identity law}) \\
 &\Leftrightarrow (\sim p \wedge (q \vee \sim q)) \vee (q \wedge (p \vee \sim p)) && (\because \text{Negation law}) \\
 &\Leftrightarrow (\sim p \wedge q) \vee (\sim p \wedge \sim q) \vee (q \wedge p) \vee (q \wedge \sim p) && (\because \text{Distributive law}) \\
 &\Leftrightarrow (\sim p \wedge q) \vee (\sim p \wedge \sim q) \vee (p \wedge q) && (\because \text{Idempotent \& commutative laws})
 \end{aligned}$$

which is required PDNF

Truth Table Method

p	q	$\sim p$	$\sim p \vee q$
T	T	F	T ✓
T	F	F	F
F	T	T	T ✓
F	F	T	T ✓

($\sim p \vee q$ True values corresponding p and q values are the min terms)

The min terms are $(p \wedge q), (\sim p \wedge q), (\sim p \wedge \sim q)$

The required PDNF is $(p \wedge q) \vee (\sim p \wedge q) \vee (\sim p \wedge \sim q)$

2. Obtain the PDNF of the $(p \wedge q) \vee (\sim p \wedge r) \vee (q \wedge r)$

Sol: $(p \wedge q) \vee (\sim p \wedge r) \vee (q \wedge r)$

$$\Leftrightarrow (p \wedge q \wedge T) \vee (\sim p \wedge r \wedge T) \vee (q \wedge r \wedge T) \quad (\because \text{Identity law})$$

$$\Leftrightarrow (p \wedge q \wedge (r \vee \sim r)) \vee (\sim p \wedge r \wedge (q \vee \sim q)) \vee (q \wedge r \wedge (p \vee \sim p)) \quad (\because \text{Negation law})$$

$$\Leftrightarrow (p \wedge q \wedge r) \vee (p \wedge q \wedge \sim r) \vee (\sim p \wedge r \wedge q) \vee (\sim p \wedge r \wedge \sim q) \vee (q \wedge r \wedge p) \vee (q \wedge r \wedge \sim p) \quad (\because \text{Distributive law})$$

$$\Leftrightarrow (p \wedge q \wedge r) \vee (p \wedge q \wedge \sim r) \vee (\sim p \wedge r \wedge q) \vee (\sim p \wedge r \wedge \sim q) \quad (\because \text{Idempotent \& commutative laws})$$

which is the required PDNF

**
* 3. Find the PDNF of $p \rightarrow [(p \rightarrow q) \wedge \sim(\sim q \wedge \sim p)]$

Sol: $p \rightarrow [(p \rightarrow q) \wedge \sim(\sim q \wedge \sim p)]$

$$\Leftrightarrow p \rightarrow [(\sim p \vee q) \wedge (q \wedge p)] \quad (\because p \rightarrow q \equiv \sim p \vee q, \text{Demorgan's law})$$

$$\Leftrightarrow p \rightarrow [\sim p \wedge (q \wedge p) \vee q \wedge (q \wedge p)] \quad (\because \text{Distributive law})$$

$$\Leftrightarrow p \rightarrow [\sim p \wedge p \wedge q \vee (q \wedge q) \wedge p] \quad (\because \text{Commutative \& Associative law})$$

$$\Leftrightarrow p \rightarrow [(F \wedge q) \vee (q \wedge p)] \quad (\because \text{Negation \& Idempotent law})$$

$$\Leftrightarrow p \rightarrow [F \vee (q \wedge p)] \quad (\because \text{Domination law})$$

$$\Leftrightarrow p \rightarrow (q \wedge p) \quad (\because \text{Identity law})$$

$$\Leftrightarrow \sim p \vee (q \wedge p) \quad (\because p \rightarrow q \equiv \sim p \vee q)$$

$$\Leftrightarrow (\sim p \wedge T) \vee (q \wedge p) \quad (\because \text{Identity law})$$

$$\Leftrightarrow (\sim p \wedge (q \vee \sim q)) \vee (q \wedge p) \quad (\because \text{Negation law})$$

$$\Leftrightarrow (\sim p \wedge q) \vee (\sim p \wedge \sim q) \vee (p \wedge q) \quad (\because \text{Distributive law})$$

which is the required PDNF

4. Obtain the PDNF of $\sim [p \vee (\sim p \wedge \sim q \wedge r)]$

sol: $\sim [p \vee (\sim p \wedge \sim q \wedge r)]$

$$\Leftrightarrow [\sim p \wedge \sim (\sim p \wedge \sim q \wedge r)] \quad (\because \text{De Morgan's law})$$

$$\Leftrightarrow [\sim p \wedge (p \vee q \vee \sim r)] \quad (\because \text{" " " "})$$

$$\Leftrightarrow (\sim p \wedge p) \vee (\sim p \wedge q) \vee (\sim p \wedge \sim r) \quad (\because \text{Distributive law})$$

$$\Leftrightarrow F \vee (\sim p \wedge q) \vee (\sim p \wedge \sim r) \quad (\because \text{Negation law})$$

$$\Leftrightarrow (\sim p \wedge q) \vee (\sim p \wedge \sim r) \quad (\because \text{Identity law})$$

$$\Leftrightarrow (\sim p \wedge q \wedge T) \vee (\sim p \wedge \sim r \wedge T) \quad (\because \text{Identity law})$$

$$\Leftrightarrow ((\sim p \wedge q) \wedge (r \vee \sim r)) \vee ((\sim p \wedge \sim r) \wedge (q \vee \sim q)) \quad (\because \text{Negation law})$$

$$\Leftrightarrow (\sim p \wedge q \wedge r) \vee (\sim p \wedge q \wedge \sim r) \vee (\sim p \wedge \sim r \wedge q) \vee (\sim p \wedge \sim r \wedge \sim q) \quad (\because \text{Distributive law})$$

$$\Leftrightarrow (\sim p \wedge q \wedge r) \vee (\sim p \wedge q \wedge \sim r) \vee (\sim p \wedge \sim r \wedge q) \quad (\because \text{Idempotent \& commutative laws})$$

which is the required PDNF

5. Find the PDNF of $q \wedge (p \vee \sim q)$

sol: $q \wedge (p \vee \sim q)$

$$\Leftrightarrow (q \wedge p) \vee (q \wedge \sim q) \quad (\because \text{Distributive law})$$

$$\Leftrightarrow (q \wedge p) \vee F \quad (\because \text{Negation law})$$

$$\Leftrightarrow (q \wedge p) \quad (\because \text{Identity law})$$

$$\Leftrightarrow p \wedge q \quad (\because \text{Commutative law})$$

which is the required PDNF

PCNF (Principle Conjunctive Normal Form)

A statement formula which is equivalent to given formula consisting of conjunction of max terms only, which is called PCNF or Product of sum canonical form.

NOTE: Max terms

2 variable terms

$$p \vee q$$

$$\sim p \vee q$$

$$p \vee \sim q$$

$$\sim p \vee \sim q$$

3 variable terms

$$p \vee q \vee r$$

$$\sim p \vee q \vee r$$

$$p \vee \sim q \vee r$$

$$p \vee q \vee \sim r$$

$$\sim p \vee \sim q \vee r$$

$$p \vee \sim q \vee \sim r$$

$$\sim p \vee q \vee \sim r$$

$$\sim p \vee \sim q \vee \sim r$$

1. Find the PCNF of $\sim(p \vee q)$

$$\begin{aligned}
 \text{sol: } \sim(p \vee q) &\Leftrightarrow \sim p \wedge \sim q & (\because \text{Demorgan's law}) \\
 &\Leftrightarrow (\sim p \vee F) \wedge (\sim q \vee F) & (\because \text{Identity law}) \\
 &\Leftrightarrow (\sim p \vee (q \wedge \sim q)) \wedge (\sim q \vee (p \wedge \sim p)) & (\because \text{Negation law}) \\
 &\Leftrightarrow (\sim p \vee q) \wedge (\sim p \vee \sim q) \wedge (\sim q \vee p) \wedge (\sim q \vee \sim p) & (\because \text{Distributive law}) \\
 &\Leftrightarrow (\sim p \vee q) \wedge (\sim p \vee \sim q) \wedge (p \vee \sim q) & (\because \text{Commutative \& Idempotent})
 \end{aligned}$$

which is the required PCNF

2. Find the PCNF of $\sim(p \rightarrow q)$

$$\begin{aligned}
 \text{sol: } \sim(p \rightarrow q) &\Leftrightarrow \sim(\sim p \vee q) & (\because p \rightarrow q \equiv \sim p \vee q) \\
 &\Leftrightarrow (p \wedge \sim q) & (\because \text{Demorgan's law}) \\
 &\Leftrightarrow (p \vee F) \wedge (\sim q \vee F) & (\because \text{Identity law}) \\
 &\Leftrightarrow (p \vee (q \wedge \sim q)) \wedge (\sim q \vee (p \wedge \sim p)) & (\because \text{Negation law}) \\
 &\Leftrightarrow (p \vee q) \wedge (p \vee \sim q) \wedge (\sim q \vee p) \wedge (\sim q \vee \sim p) & (\because \text{Distributive law}) \\
 &\Leftrightarrow (p \vee q) \wedge (p \vee \sim q) \wedge (\sim p \vee \sim q) & (\because \text{Commutative \& Idempotent})
 \end{aligned}$$

which is the required PCNF

3. Obtain the PCNF of the following:

i. $\sim(p \leftrightarrow q)$

ii. $(\sim p \rightarrow r) \wedge (q \leftrightarrow p)$

iii. $(\sim p \rightarrow r) \wedge (q \leftrightarrow p)$

$$\begin{aligned}
 &\Leftrightarrow (p \vee r) \wedge [(q \rightarrow p) \wedge (p \rightarrow q)] & \left[\begin{array}{l} p \rightarrow q \equiv \sim p \vee q \\ p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \end{array} \right] \\
 &\Leftrightarrow (p \vee r) \wedge (\sim q \vee p) \wedge (\sim p \vee q) & (\because p \rightarrow q \equiv \sim p \vee q) \\
 &\Leftrightarrow ((p \vee r) \vee F) \wedge ((\sim q \vee p) \vee F) \wedge ((\sim p \vee q) \vee F) & (\because \text{Identity law}) \\
 &\Leftrightarrow [(p \vee r) \vee (q \wedge \sim q)] \wedge [(p \vee r) \vee (p \wedge \sim p)] \wedge [(\sim p \vee q) \vee (r \wedge \sim r)] & (\because \text{Negation law}) \\
 &\Leftrightarrow (p \vee r \vee q) \wedge (p \vee r \vee \sim q) \wedge (\sim q \vee p \vee r) \wedge (\sim q \vee p \vee \sim r) \wedge (\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r) & (\because \text{Distributive law}) \\
 &\Leftrightarrow (p \vee q \vee r) \wedge (p \vee \sim q \vee r) \wedge (p \vee q \vee \sim r) \wedge (\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r) & \\
 &\Leftrightarrow (p \vee q \vee r) \wedge (\sim p \vee q \vee r) \wedge (p \vee \sim q \vee r) \wedge (p \vee q \vee \sim r) \wedge (\sim p \vee q \vee \sim r) & \\
 & & (\because \text{commutative \& Idempotent laws})
 \end{aligned}$$

which is the required PCNF

$$3. \sim (p \leftrightarrow q)$$

$$\Leftrightarrow \sim [(p \rightarrow q) \wedge (q \rightarrow p)]$$

$$\Leftrightarrow \sim [(\sim p \vee q) \wedge (\sim q \vee p)]$$

$$\Leftrightarrow [\sim (\sim p \vee q)] \vee [\sim (\sim q \vee p)]$$

$$\Leftrightarrow (p \wedge \sim q) \vee (q \wedge \sim p)$$

$$X(\Leftrightarrow ((p \wedge \sim q) \vee F) \vee ((q \wedge \sim p) \vee F))$$

$$\Leftrightarrow (p \wedge \sim q) \vee (q \wedge \sim p)$$

$$\Leftrightarrow (p \wedge \sim q) \vee (q \wedge \sim p)$$

$$4. \sim (p \leftrightarrow q)$$

$$\Leftrightarrow \sim [(p \rightarrow q) \wedge (q \rightarrow p)]$$

$$(\because p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p))$$

$$\Leftrightarrow \sim [(\sim p \vee q) \wedge (\sim q \vee p)]$$

$$(\because p \rightarrow q \equiv \sim p \vee q)$$

$$\Leftrightarrow (p \wedge \sim q) \vee (q \wedge \sim p)$$

$$(\because \text{Demorgan's law})$$

$$\Leftrightarrow (p \vee (q \wedge \sim p)) \wedge (\sim q \vee (q \wedge \sim p))$$

$$(\because \text{Distributive law})$$

$$\Leftrightarrow (p \vee q) \wedge (p \vee \sim p) \wedge (\sim q \vee q) \wedge (\sim q \vee \sim p)$$

$$(\because \text{Distributive law})$$

$$\Leftrightarrow (p \vee q) \wedge T \wedge T \wedge (\sim q \vee \sim p)$$

$$(\because \text{Negation law})$$

$$\Leftrightarrow (p \vee q) \wedge (\sim p \vee \sim q)$$

$$(\because \text{Identity law})$$

$$\Leftrightarrow (p \vee q) \wedge (\sim p \vee \sim q)$$

$$(\because \text{commutative law})$$

which is the required PCNF

4. Find the PCNF of $(p \rightarrow q) \wedge (q \leftrightarrow r)$

$$\text{sol: } (p \rightarrow q) \wedge (q \leftrightarrow r)$$

$$\Leftrightarrow (\sim p \vee q) \wedge [(q \rightarrow r) \wedge (r \rightarrow q)]$$

$$(\because p \rightarrow q \equiv \sim p \vee q \text{ \& } p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p))$$

$$\Leftrightarrow (\sim p \vee q) \wedge [(q \rightarrow r) \wedge (\sim r \vee q)]$$

$$(\because p \rightarrow q \equiv \sim p \vee q)$$

$$\Leftrightarrow [(\sim p \vee q) \vee F] \wedge [(q \rightarrow r) \vee F] \wedge [(\sim r \vee q) \vee F]$$

$$(\because \text{Identity law})$$

$$\Leftrightarrow [(\sim p \vee q) \vee (r \wedge \sim r)] \wedge [(\sim q \vee r) \vee (p \wedge \sim p)] \wedge [(\sim r \vee q) \vee (p \wedge \sim p)]$$

$$(\because \text{Negation law})$$

$$\Leftrightarrow (\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r) \wedge (\sim q \vee r \vee p) \wedge (\sim q \vee r \vee \sim p) \wedge (\sim r \vee q \vee p) \wedge (\sim r \vee q \vee \sim p)$$

$$(\because \text{Distributive law})$$

$$\Leftrightarrow (\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r) \wedge (p \vee \sim q \vee r) \wedge (\sim p \vee \sim q \vee r) \wedge (p \vee q \vee \sim r)$$

$$(\because \text{Idempotent \& commutative laws})$$

which is the required PCNF

Truth table method

p	q	r	$p \rightarrow q$	$q \leftrightarrow r$	$(p \rightarrow q) \wedge q \leftrightarrow r$
T	T	T	T	T	T
T	T	F	F	F	F ✓
T	F	T	T	F	F ✓
T	F	F	F	T	F ✓
F	T	T	T	T	T
F	T	F	T	F	F ✓
F	F	T	T	F	F ✓
F	F	F	T	T	T

The max terms are

$$(p \vee q \vee \sim r) \wedge (p \vee \sim q \vee \sim r) \wedge (p \vee \sim q \vee r) \wedge (\sim p \vee q \vee \sim r) \wedge (\sim p \vee \sim q \vee r)$$

The required PCNF is

$$(p \vee q \vee \sim r) \wedge (p \vee \sim q \vee r) \wedge (p \vee \sim q \vee \sim r) \wedge (\sim p \vee q \vee \sim r) \wedge (\sim p \vee \sim q \vee r)$$

5. Find the PCNF of the $(\sim p \vee \sim q) \rightarrow (p \leftrightarrow \sim q)$

Sol $(\sim p \vee \sim q) \rightarrow (p \leftrightarrow \sim q)$

$$\Leftrightarrow \sim(\sim p \vee \sim q) \vee (p \leftrightarrow \sim q)$$

$$\Leftrightarrow \sim(\sim p \vee \sim q) \vee (p \leftrightarrow \sim q) \quad (\because p \leftrightarrow q \equiv \sim p \vee q)$$

$$\Leftrightarrow (p \wedge q) \vee (p \leftrightarrow \sim q) \quad (\because \text{Demorgan's law})$$

$$\Leftrightarrow (p \wedge q) \vee [(p \rightarrow \sim q) \wedge (\sim q \rightarrow p)] \quad [\because p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)]$$

$$\Leftrightarrow (p \wedge q) \vee [(\sim p \vee \sim q) \wedge (q \vee p)] \quad (\because p \rightarrow q \equiv \sim p \vee q)$$

$$\Leftrightarrow (p \wedge q) \vee (\sim p \wedge q \vee \sim p \wedge p \vee \sim q \wedge q \vee \sim q \wedge p)$$

$$\Leftrightarrow (p \wedge q) \vee (\sim p \wedge q \vee \sim q \wedge p)$$

$$\Leftrightarrow (p \wedge q) \vee [(\sim p \wedge q \vee p) \vee (\sim q \wedge q \vee p)] \quad (\because \text{Distributive law})$$

$$\Leftrightarrow (p \wedge q) \vee [(\sim p \vee p) \wedge q]$$

Theory of inference:

We can divide the new statement from the old statement by using rules of inference is called inference.

1.
$$\begin{array}{c} \text{premises} \\ \text{or} \\ \text{Axioms} \end{array} \quad \text{or hypotheses}$$

$$\text{conclusion}$$

2.
$$\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_n \end{array} \left. \vphantom{\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_n \end{array}} \right\} \rightarrow \text{premises}$$

$$C \left. \vphantom{C} \right\} \rightarrow \text{conclusion}$$

3.
$$\frac{P_1 \wedge P_2 \wedge P_3 \dots \wedge P_n}{C} \rightarrow \text{conjunction of premises}$$

$$C \rightarrow \text{conclusion}$$

$$[P_1 \wedge P_2 \wedge P_3 \dots \wedge P_n] \rightarrow C$$

Conjunction of premises conditional conclusion

Rules of inference:

Tautological form

Name.

1.
$$\frac{P \wedge q}{\therefore P} \text{ (or)} \frac{P \wedge q}{\therefore q}$$

$$(P \wedge q) \rightarrow P \text{ (or)} \\ (P \wedge q) \rightarrow q$$

Simplification

2.
$$\frac{P}{\therefore P \vee q} \text{ (or)} \frac{q}{\therefore P \vee q}$$

$$P \rightarrow (P \vee q) \text{ (or)} \\ q \rightarrow (P \vee q)$$

Addition

3.
$$\frac{P}{\therefore P \wedge q} \quad \frac{q}{\therefore P \wedge q}$$

$$(P \wedge q) \rightarrow (P \wedge q)$$

Conjunction

4.
$$\frac{P \rightarrow q}{\therefore q} \quad \frac{P}{\therefore q}$$

$$[(P \rightarrow q) \wedge P] \rightarrow q$$

Modus ponens

5.
$$\frac{P \rightarrow q}{\therefore \sim p} \quad \frac{\sim q}{\therefore \sim p}$$

$$[(P \rightarrow q) \wedge (\sim q)] \rightarrow \sim p$$

Modus ~~Tonens~~ Tonens

6.
$$\frac{P \rightarrow q}{\therefore P \rightarrow r} \quad \frac{q \rightarrow r}{\therefore P \rightarrow r}$$

$$[(P \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (P \rightarrow r)$$

Hypothetical sylloism
(or)

Transitive rule

Rule	Autological form	Name
1. $\frac{p \vee q \quad \sim p}{\therefore q}$	$[(p \vee q) \wedge (\sim p)] \rightarrow q$	Disjunction sylloism
8. $\frac{(p \rightarrow q) \wedge (r \rightarrow s) \quad p \vee r}{\therefore q \vee s}$	$[(p \rightarrow q) \wedge (r \rightarrow s)] \wedge (p \vee r) \rightarrow q \vee s$	Constructive Conjunction Delemma
9. $\frac{(p \rightarrow q) \wedge (r \rightarrow s) \quad (\sim q \wedge \sim s)}{\therefore \sim p \vee \sim r}$	$[(p \rightarrow q) \wedge (r \rightarrow s)] \wedge (\sim q \wedge \sim s) \rightarrow (\sim p \vee \sim r)$	Destructive Delemma

Note:

1. The converse of $p \rightarrow q$ is $q \rightarrow p$
2. The Contrapositive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$
3. The inverse of $p \rightarrow q$ is $\sim p \rightarrow \sim q$

Note:

1. $p \rightarrow q \equiv \sim p \vee q$
2. $\sim p \rightarrow q \equiv p \vee q$
3. $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
 $\equiv (p \wedge q) \vee (\sim q \wedge \sim p)$

Type-1:

1. Demonstrate that r is a valid inference from the premises $p \rightarrow q$, $q \rightarrow r$ and p

Sol Premises:

$p \rightarrow q$, $q \rightarrow r$, p

Conclusion:

{1}	(1)	$p \rightarrow q$	Rule p
{2}	(2)	$q \rightarrow r$	Rule p
{1,2}	(3)	$p \rightarrow r$	Hypothetical Sylloism
{4}	(4)	p	Rule p
{1,2,4}	(5)	r	Modus ponens

2. Show that $\sim q, p \rightarrow q \Rightarrow \sim p$

Sol: Premises:

$$\sim q, p \rightarrow q$$

Conclusion:

$$\sim p$$

{1}	(1)	$p \rightarrow q$	Rule p
{2}	(2)	$\sim q \rightarrow \sim p$	Contrapositive
{3}	(3)	$\sim q$	Rule p
{1,3}	(4)	$\sim p$	Modus Ponens

3. Show that (rvs) follows logically from the premises (cvd), $(cvd) \rightarrow \sim h$, $\sim h \rightarrow (a \wedge \sim b)$, $(a \wedge \sim b) \rightarrow (rvs)$

Sol: Premises:

$$(cvd), (cvd) \rightarrow \sim h, \sim h \rightarrow (a \wedge \sim b), (a \wedge \sim b) \rightarrow (rvs)$$

Conclusion:

$$(rvs)$$

{1}	(1)	(cvd)	Rule p
{2}	(2)	$(cvd) \rightarrow \sim h$	Rule p
{1,2}	(3)	$\sim h$	Modus ponens
{4}	(4)	$\sim h \rightarrow (a \wedge \sim b)$	Rule p
{1,2,4}	(5)	$a \wedge \sim b$	Modus ponens
{6}	(6)	$(a \wedge \sim b) \rightarrow (rvs)$	Rule p
{1,2,4,6}	(7)	rvs	Modus ponens

4. Show that svr is tautologically implied by $(pvq) \wedge (p \rightarrow r) \wedge (q \rightarrow s)$

Sol: Given that

Premises:

$$pvq, p \rightarrow r, q \rightarrow s$$

Conclusion:

$$svr$$

{1}	(1)	pvq	Rule p
{1}	(2)	$\sim p \rightarrow q$	$p \rightarrow q \equiv \sim p \vee q$
{1}	(3)	$\sim q \rightarrow p$	Contrapositive
{4}	(4)	$p \rightarrow r$	Rule p

{1,4}	(5)	$\sim q \rightarrow r$	Hypothetical Sylloism
{1,4}	(6)	$\sim r \rightarrow q$	Contrapositive
{1,4}	(7)	$q \rightarrow s$	Rule p
{1,4,7}	(8)	$\sim r \rightarrow s$	Hypothetical Sylloism
{1,4,7}	(9)	$r \vee s$	$p \rightarrow q \equiv \sim p \vee q$
{1,4,7}	(10)	$s \vee r$	Commutative law

5. Show that $R \wedge (P \vee Q)$ is a valid conclusion from the premises $(P \vee Q)$, $(Q \rightarrow R)$, $(P \rightarrow M)$ and $\sim M$

solⁿ Given that
Premises:

$(P \vee Q)$, $(Q \rightarrow R)$, $(P \rightarrow M)$, $\sim M$

Conclusion

$R \wedge (P \vee Q)$

{1}	(1)	$P \vee Q$	Rule p
{1}	(2)	$\sim P \rightarrow Q$	$P \rightarrow Q \equiv \sim P \vee Q$
{3}	(3)	$Q \rightarrow R$	Rule p
{1,3}	(4)	$\sim P \rightarrow R$	Hypothetical Sylloism
{1,3,4}	(5)	$\sim R \rightarrow P$	Contrapositive
{6}	(6)	$P \rightarrow M$	Rule p
{1,3,6}	(7)	$\sim R \rightarrow M$	Hypothetical Sylloism
{1,3,6,7}	(8)	$\sim M \rightarrow R$	Contrapositive
{9}	(9)	$\sim M$	Rule p
{1,3,6,9}	(10)	R	Modus Ponens
{1,3,6,9}	(11)	$R \wedge (P \vee Q)$	Conjunction

Type-2:

1. Test the validity of the following argument: "Sonia is watching TV. If Sonia is watching TV, then she is not studying. If she is not studying, then her father will not buy a scooter. Therefore, Sonia's father will not buy a scooter."

Sol: Let us indicate the statements as follows

P : Sonia is watching TV

Q : Sonia is studying

r : Sonia's father will buy a scooter

Premises:

$P, P \rightarrow \sim Q, \sim Q \rightarrow \sim r$

Conclusion:

$\sim r$

{1}	(1)	P	Rule p
{2}	(2)	$P \rightarrow \sim Q$	Rule p
{1,2}	(3)	$\sim Q$	Modus ponens
{4}	(4)	$\sim Q \rightarrow \sim r$	Rule p
{1,2,4}	(5)	$\sim r$	Modus ponens

2. "If there was a ball game, then travelling was difficult. If they arrived on time, then travelling was not difficult. They arrived on time. Therefore, there was no ball game". Show that these statements constitute a valid argument

Sol: Let us indicate the statements as follows:

P : There was a ball game

Q : Travelling was difficult

r : They arrived on time

Premises:

$(P \rightarrow Q), (r \rightarrow \sim Q), r,$

Conclusion:

$\sim P$

{1}	(1)	$P \rightarrow Q$	Rule p
{2}	(2)	$Q \rightarrow \sim r$	Contrapositive
{3}	(3)	$r \rightarrow \sim Q$	Rule p

watching
dying.
buy a
"

{2}	(3)	$q \rightarrow \sim r$	contrapositive
{1,3}	(4)	$p \rightarrow \sim r$	hypothetical sylloism
{5}	(5)	r	Rule p
{1,3,5}	(6)	$\sim p$	Modus Tonens

3. "If clifton does not live in france then he does not speak french. clifton does not drive a datsun. If clifton lives in france, then he rides a bicycle. Either clifton speaks french or he drives a datsun. Hence, clifton rides a bicycle"

sol Let us indicate the statements as follows

p = clifton live in france

q = clifton speaks french

r = clifton drives a datsun

s = clifton rides a bicycle

Premises:

$(\sim p \rightarrow \sim q), \sim r, (p \rightarrow s), q \vee r$

conclusion:

s

{1}	(1)	$\sim p \rightarrow \sim q$	Rule p
{2}	(2)	$q \rightarrow p$	contrapositive
{3}	(3)	$p \rightarrow s$	Rule p
{1,3}	(4)	$q \rightarrow s$	Hypothetical Sylloism
{1,3}	(5)	$\sim s \rightarrow \sim q$	Contrapositive
{6}	(6)	$q \vee r$	Rule p
{6}	(7)	$\sim q \rightarrow r$	$p \rightarrow q \equiv \sim p \vee r$
{1,3,6}	(8)	$\sim s \rightarrow r$	Hypothetical Sylloism
{1,3,6}	(9)	$\sim r \rightarrow s$	Contrapositive
{10}	(10)	$\sim r$	Rule p
{1,3,6,10}	(11)	s	Modus ponens

Type-3:

* Rule CP (Conditional Proof)

If we can derive S from R and set of premises, then we can derive $R \rightarrow S$ from a set of premises alone

Note:

The conclusion is of the form $p \rightarrow q$ then p is taken to additional premises or assumed premises

1. Show that $r \rightarrow s$ can be derived from the premises $p \rightarrow (q \rightarrow s)$, $(\neg r \vee p)$ and q

Sol Premises:

$p \rightarrow (q \rightarrow s)$, $(\neg r \vee p)$, q

Conclusion

$r \rightarrow s$

{1}	(1)	r	Rule P (Assumed premises)
{2}	(2)	$\neg r \vee p$	Rule P
{1,2}	(3)	$r \rightarrow p$	$p \rightarrow q \equiv \neg p \vee q$
{1,2}	(4)	p	Modus ponens
{5}	(5)	$p \rightarrow (q \rightarrow s)$	Rule P
{1,2,5}	(6)	$q \rightarrow s$	Modus Ponens
{7}	(7)	q	Rule P
{1,2,5,7}	(8)	s	Modus Ponens
{1,2,5,7}	(9)	$r \rightarrow s$	Rule CP

2. By using the method of derivation, show that the following statements constitute a valid argument: "If A works hard, then either B or C will enjoy. If B enjoys, then A will not work hard. If D enjoys, then C will not. Therefore, If A works hard, D will not enjoy."

Sol Let us indicate the statements as follows:

p : A works hard

q : B will enjoy

r : C will enjoy

s : D will enjoy

Premises:

Conclusion:

$p \rightarrow \sim s$

{1}	(1)	p	Rule p (Assumed premises)
{2}	(2)	$p \rightarrow (q \vee r)$	Rule p
{1,2}	(3)	$q \vee r$	Modus ponens
{1,2}	(4)	$\sim q \rightarrow r$	$p \rightarrow q \equiv \sim q \vee r$
{1,2}	(5)	$\sim r \rightarrow q$	Contrapositive
{6}	(6)	$q \rightarrow \sim p$	Rule p
{1,2,6}	(7)	$\sim r \rightarrow \sim p$	Hypothetical Syllogism
{1,2,6}	(8)	$\sim p \rightarrow \sim r$	Converse
{9}	(9)	$s \rightarrow \sim r$	Rule p
{9}	(10)	$\sim r \rightarrow s$	converse
	(11)	$r \rightarrow \sim s$	Inverse
{1,2,6,9}	(11)	$\sim p \rightarrow s$	Hypothetical Syllogism
{1,2,6,9}	(12)	$p \rightarrow \sim s$	Inverse

Optimal step

{1,2,6,9}	(13)	$\sim s$ [(1), (12)]	modus ponens
{1,2,6,9}	(14)	$p \rightarrow \sim s$ [(1), (13)]	Rule CP

Type 4:

Indirect method of proof:

1. Show that $\sim(p \wedge q)$ follows from $\sim p \wedge \sim q$

2. Given that

Premises:

$\sim p \wedge \sim q$

Conclusion:

$\sim(p \wedge q)$

{1}	(1)	$\sim(\sim(p \wedge q))$	Rule p
{1}	(2)	$p \wedge q$	Double negation law
{1}	(3)	p	Simplification
{1}	(4)	q	Simplification
{5}	(5)	$\sim p \wedge \sim q$	Rule p
{5}	(6)	$\sim p$	Simplification
{5}	(7)	$\sim q$	Simplification

{1,5} (2) $p \wedge \sim p$ Conjunction

$$\therefore p \wedge \sim p \equiv F$$

$\therefore$ Our assumption is wrong

$\therefore \sim(p \wedge q)$ follows from $\sim p \wedge \sim q$

2. Show that $p \rightarrow q, q \rightarrow r, \sim(p \wedge r), p \vee r \leftrightarrow r$ by using the indirect method

Sol: Given that

Premises:

$$p \rightarrow q, q \rightarrow r, \sim(p \wedge r), p \vee r$$

Conclusion:

r

{1}	(1)	$\sim r$	Rule p
{2}	(2)	$q \rightarrow r$	Rule p
{1,2}	(3)	$\sim q$	Modus Tollens
{4}	(4)	$p \rightarrow q$	Rule p
{1,2,4}	(5)	$\sim p$	Modus Tollens
{6}	(6)	$p \vee r$	Rule p
{6}	(7)	$\sim p \rightarrow r$	$p \rightarrow q \equiv \sim p \vee q$
{1,2,4,6}	(8)	r	Modus Ponens
{1,2,4,6}	(9)	$\sim r \wedge r$	Conjunction

$$\therefore r \wedge \sim r \equiv F$$

$\therefore$ Our assumption is wrong

$\therefore p \rightarrow q, q \rightarrow r, \sim(p \wedge r), p \vee r \leftrightarrow r$

Type 5:

Inconsistent

1. Show that the premises $a \rightarrow (b \rightarrow c), d \rightarrow (b \wedge \sim c)$, and are inconsistent

Sol: Given that

premises:

$$a \rightarrow (b \rightarrow c), d \rightarrow (b \wedge \sim c), \text{ and}$$

{1}	(1)	and	Rule p
{1}	(2)	a	Simplification
{1}	(3)	d	Simplification
{4}	(4)	$a \rightarrow (b \rightarrow c)$	Rule p

{1,4}	(5)	$b \rightarrow c$	Modus Ponens
{1,4}	(6)	$\sim b \vee c$	$p \rightarrow q \equiv \sim p \vee q$
{1,4}	(7)	$\sim(b \wedge c)$	Demorgan's law
{8}	(8)	$d \rightarrow (b \wedge c)$	Rule p
{1,4,8}	(9)	and	Modus Ponens
{1,4,8}	(10)	$\sim d \wedge d$	Conjunction

$$\therefore \sim d \wedge d \equiv F$$

$\therefore$ Given premises are Inconsistent

2. Show that the following set of premises is Inconsistent:

"If the contract is valid, then John is liable for penalty. If John is liable for penalty, he will go bankrupt. If the bank will loan him money, he will not go bankrupt. As a matter of fact, the contract is valid and the bank will loan him money".

sol: Let us indicate the statements as follows:

P: The contract is valid

Q: John is liable for penalty

R: John will go bankrupt

S: The bank will loan him money

Premises:

$$p \rightarrow q, q \rightarrow r, s \rightarrow \sim r, p \wedge s$$

{1}	(1)	$p \rightarrow q$	Rule p
{2}	(2)	$q \rightarrow r$	Rule p
{1,2}	(3)	$p \rightarrow r$	Hypothetical Sylloism
{4}	(4)	$s \rightarrow \sim r$	Rule p
{4}	(5)	$r \rightarrow \sim s$	Contrapositive
{1,2,4}	(6)	$p \rightarrow \sim s$	Hypothetical Sylloism
{1,2,4}	(7)	$\sim p \vee \sim s$	$p \rightarrow q \equiv \sim p \vee q$
{1,2,4}	(8)	$\sim(p \wedge s)$	Demorgan's law
{9}	(9)	$(p \wedge s)$	Rule p
{1,2,4,9}	(10)	$\sim(p \wedge s) \wedge (p \wedge s)$	Conjunction

$$\therefore \sim(p \wedge s) \wedge (p \wedge s) \equiv F$$

$\therefore$ The given premises are Inconsistent

3. Show that the following premises are inconsistent:

1. If Jack misses many classes because of illness, then he fails highschool
2. Jack fails highschool, then he is uneducated
3. If Jack reads a lot of books, then he is not uneducated.
4. Jack misses many classes because of illness and reads a lot of books

Solⁿ Let us indicate the statements as follows:

P : Jack misses many classes because of illness

q : Jack fails highschool

r : Jack is uneducated

s : Jack reads a lot of books

Premises:

$P \rightarrow q, q \rightarrow r, s \rightarrow \sim r, P \wedge s$

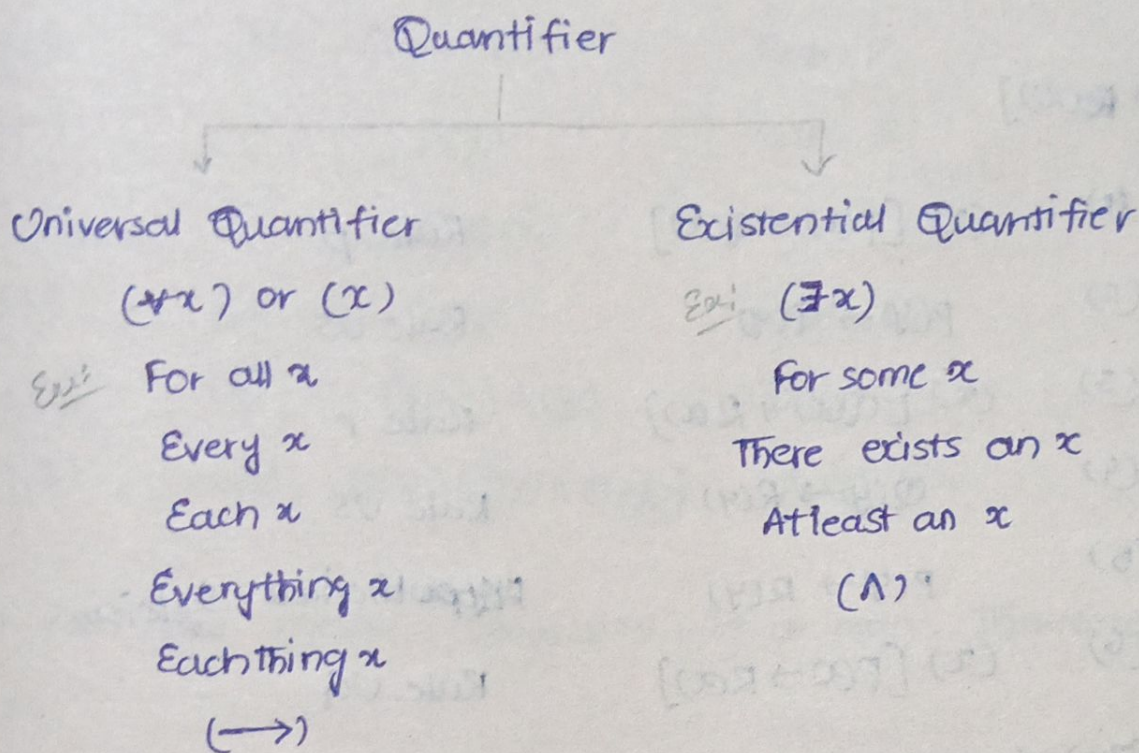
$\{1\}$	(1)	$P \rightarrow q$	Rule p
$\{2\}$	(2)	$q \rightarrow r$	Rule p
$\{1, 2\}$	(3)	$P \rightarrow r$	Hypothetical Sylloism
$\{4\}$	(4)	$s \rightarrow \sim r$	Rule p
$\{1, 4\}$	(5)	$r \rightarrow \sim s$	Contrapositive
$\{1, 2, 4\}$	(6)	$P \rightarrow \sim s$	Hypothetical Sylloism

Quantifiers:

The phrase which indicates the quantity is called Quantifier.

Quantifiers are two types:

1. Universal Quantifier
2. Existential Quantifier



Rules of Quantifiers:

Rule US: Universal Specification R_u

$$x A(x) \Rightarrow A(y) \quad \text{all to one}$$

Rule ES: Existential Specification

$$(\exists x) A(x) \Rightarrow A(y) \quad \text{some to one}$$

Rule UG: Universal Generalization

$$A(x) \Rightarrow (\forall y) A(y) \quad \text{one to all}$$

Rule EG: Existential Generalization

$$A(x) \Rightarrow (\exists y) A(y) \quad \text{one to some}$$

Note:

$$1. (\exists x) [A(x) \vee B(x)] \equiv (\exists x) A(x) \vee (\exists x) B(x)$$

$$2. (x) [A(x) \wedge B(x)] \equiv (x) A(x) \wedge (x) B(x)$$

$$3. \sim (x) A(x) \equiv (\exists x) \sim A(x)$$

$$4. \sim (\exists x) A(x) \equiv (x) \sim A(x)$$

$$5. \sim (p \rightarrow q) \equiv p \wedge \sim q$$

Type 1:

1. Show that $(x) [P(x) \rightarrow Q(x)] \wedge (x) [Q(x) \rightarrow R(x)] \Rightarrow (x) [P(x) \rightarrow R(x)]$

Sol: Given that

Premises:

$(x) [P(x) \rightarrow Q(x)]$, $(x) [Q(x) \rightarrow R(x)]$

Conclusion:

$(x) [P(x) \rightarrow R(x)]$

{1}	(1)	$(x) [P(x) \rightarrow Q(x)]$	Rule P
{1}	(2)	$P(y) \rightarrow Q(y)$	Rule US
{3}	(3)	$(x) [Q(x) \rightarrow R(x)]$	Rule P
{3}	(4)	$Q(y) \rightarrow R(y)$	Rule US
{1, 3}	(5)	$P(y) \rightarrow R(y)$	Hypothetical Sylloism
{1, 3}	(6)	$(x) [P(x) \rightarrow R(x)]$	Rule UG

2. Show that $(\exists x) M(x)$ follows logically from the premises $(x) [H(x) \rightarrow M(x)]$ and $(\exists x) H(x)$

3. Show that $(\exists x) [P(x) \wedge Q(x)] \Rightarrow (\exists x) P(x) \wedge (\exists x) Q(x)$

2. Sol: Given that

Premises:

$(x) [H(x) \rightarrow M(x)]$, $(\exists x) H(x)$

Conclusion:

$(\exists x) M(x)$

{1}	(1)	$(x) [H(x) \rightarrow M(x)]$	Rule P
{1}	(2)	$H(y) \rightarrow M(y)$	Rule US
{3}	(3)	$\exists x H(x)$	Rule P
{3}	(4)	$H(y)$	Rule ES
{1, 3}	(5)	$M(y)$	Modus Ponens
{1, 3}	(6)	$(\exists x) M(x)$	Rule EG

$\rightarrow R(x)$

sol:

Given that

Premises:

$(\exists x) P(x), (\exists x) Q(x)$

Conjunction:

$(\exists x) [P(x) \wedge Q(x)]$

{1}	(1)	$(\exists x) P(x)$	Rule P
{1}	(2)	$P(y)$	Rule ES
{3}	(3)	$(\exists x) Q(x)$	Rule P
{3}	(4)	$Q(y)$	Rule ES
{1, 3}	(5)	$P(y) \wedge Q(y)$	Conjunction
{1, 3}	(6)	$(\exists x) [P(x) \wedge Q(x)]$	Rule EG

* "All men are mortal. Socrates is a man. Therefore

Type-2:

1. Verify the validity of the following argument:

"All men are mortal. Socrates is a man. Therefore Socrates is mortal".

sol: let us indicate the statement as follows:

$M(x)$: x is a man

$H(x)$: x is a mortal.

s : socrates

Premises:

$(x) [M(x) \rightarrow H(x)], M(s)$

Conclusion:

$H(s)$

{1}	(1)	$(x) [M(x) \rightarrow H(x)]$	Rule p
{1}	(2)	$M(s) \rightarrow H(s)$	Rule VS
{1, 3}	(3)	$M(s)$	Rule p
{1, 3}	(4)	$H(s)$	Modus Ponens

2. verify the validity of the following argument: "Every living thing is a plant or an animal. Joe's gold fish is alive and it is not a plant. All animals have hearts. Therefore, Joe's gold fish has a heart".

sol. Let us indicate the statement as follows:

$P(x)$: x is a plant

$A(x)$: x is an animal

$H(x)$: x is a heart

f = Joe's gold fish

Premises:

$(x) [P(x) \vee A(x)], \sim P(f), (x) [A(x) \rightarrow H(x)]$

Conclusion:

$H(f)$

{13}	(1)	$(x) [P(x) \vee A(x)]$	Rule p
{13}	(2)	$P(f) \vee A(f)$	Rule US
{13}	(3)	$\sim P(f) \rightarrow A(f)$	$P \rightarrow Q \equiv \sim P \vee Q$
{43}	(4)	$\sim P(f)$	Rule p
{1,43}	(5)	$A(f)$	Modus ponens
{6}	(6)	$(x) [A(x) \rightarrow H(x)]$	Rule p
{6}	(7)	$A(f) \rightarrow H(f)$	Rule US
{1,4,6}	(8)	$H(f)$	Modus ponens

3. Establish the validity of the following argument: "All integers are ~~new~~ rational numbers. Some integers are powers of two. Therefore, some rational numbers are powers of two."

sol. Let us indicate the statements as follows:

$I(x)$: x is an integer

$R(x)$: x is a rational number

$P(x)$: x is a powers of two

Premises:

$(x) [I(x) \rightarrow R(x)], \exists(x) (I(x) \wedge P(x))$

Conclusion:

$\exists(x) [R(x) \wedge P(x)]$

{13}	(1)	$\exists(x) (I(x) \wedge P(x))$	Rule p
{13}	(2)	$I(y) \wedge P(y)$	Rule ES

living
and it
Joe's

{1}	(3)	$I(y)$	Simplification
{1}	(4)	$P(y)$	Simplification
{5}	(5)	$(x) (I(x) \rightarrow R(x))$	Rule P
{5}	(6)	$I(y) \rightarrow R(y)$	Rule US
{1,5}	(7)	$R(y)$	Modus Ponens
{1,5}	(8)	$R(y) \wedge P(y)$	Conjunction
{1,5}	(9)	$\exists(x) [R(x) \wedge P(x)]$	Rule EG

4.

"Babies are illogical. Nobody is despised who can manage a crocodile. Illogical people are despised. Hence, babies cannot manage ~~or~~ crocodile."

sol: Let us indicate the statements as follows:

$B(x)$: x is a baby

$I(x)$: x is illogical

$D(x)$: x is despised

$C(x)$: x is ~~able~~ can manage a crocodile.

Premises:

$(x) [B(x) \rightarrow I(x)]$, $(x) [C(x) \rightarrow \sim D(x)]$, $(x) [I(x) \rightarrow D(x)]$

Conclusion:

$(x) [B(x) \rightarrow \sim C(x)]$

{1}	(1)	$(x) [B(x) \rightarrow I(x)]$	Rule P
{1}	(2)	$B(y) \rightarrow I(y)$	Rule US
{3}	(3)	$(x) [I(x) \rightarrow D(x)]$	Rule P
{3}	(4)	$I(y) \rightarrow D(y)$	Rule US
{1,3}	(5)	$B(y) \rightarrow D(y)$	Hypothetical Syllogism
{6}	(6)	$(x) [C(x) \rightarrow \sim D(x)]$	Rule P
{6}	(7)	$C(y) \rightarrow \sim D(y)$	Rule US
{6}	(8)	$D(y) \rightarrow \sim C(y)$	Contrapositive
{1,3,6}	(9)	$B(y) \rightarrow \sim C(y)$	Hypothetical syllogism
{1,3,6}	(10)	$(x) [B(x) \rightarrow \sim C(x)]$	Rule UG